

Manoj Kaushik

(Member and former student chair IEEE-GRSS Kerala chapter)

[G Scholar](#) | [LinkedIn](#) | +91- 9519783489 | [Website](#) | [GitHub](#)

e-mails: manojkaushik93@gmail.com | manojkaushik.22@res.iist.ac.in

Computer Science researcher and Assistant Professor working on machine learning, deep learning and generative models for high-dimensional spectral and image data, with a focus on precision agriculture and remote sensing. Additional work in biomedical signals and NLP. Teaching base: Design and Analysis of Algorithms, Data Structures, Responsible and Explainable AI, Compiler Design.

EDUCATION

- Indian Institute of Space Science and Technology (IIST) Thiruvananthapuram (ISRO-DoS, Kerala, IN)
- Ph.D. in Computer Science (Machine Learning and Deep Learning for High-Dimensional Spectral Data) Jan'2022– present
 - Area: ML and DL (generative models, computer vision) with applications in satellite and drone remote sensing
- Centre for Advanced Studies (CAS), AKTU Lucknow (Govt. Research Institute, Uttar Pradesh, IN)
- M.Tech. in CSE-ML Specialization (Gold Medal) 2019-21
 - Master's Dissertation: Analysis and Diagnosis of Specific Language Impairment Problem Using Deep Learning
- Kamla Nehru Institute of Technology (KNIT) Sultanpur (Govt. Engineering College, Uttar Pradesh, IN)
- B.Tech. in Information Technology 2012-16
 - Project: Ease At KNIT: An intranet e-commerce web portal

TEACHING EXPERIENCE

- Assistant Professor at Graphic Era Hill University, Bhimtal Jan'2026-Present
- Courses: "Design and Analysis of Algorithms", "Data Structures", "Responsible and Explainable AI", and "Compiler Design" B.Tech level
- Teaching Assistant to Master students at IIST, Thiruvananthapuram Aug-Dec (2024 & 2025)
- Course 'Probability and Statistics for Geo-informatics'
- Teaching Assistant to Bachelor students at MMMUT, Gorakhpur Oct-Dec 2021
- Course 'Fundamentals of Machine Learning and Artificial Intelligence'

INDUSTRY EXPERIENCE

- Data Science Internship in KaleidEO (a SatSure company), Bengaluru Jan-Apr 2024
- End-to-end ML segmentation modeling of high-yield vegetable crops using multispectral drone
- Project Executive Officer in MeitY GOI and MMMUT, Gorakhpur Oct-Jan 2022
- The project title is "Development of IoT and drone-based agriculture monitoring system with the objective of skill development of a socially deprived community."
- Software Developer in Kranti Tech Services Pvt. Ltd., Noida Mar-Oct 2021
- Applied Machine learning to classify grievance emails using NLP. Used Term Frequency and Inverse Document Frequency (TF-IDF) for feature extraction from the prepared dataset.
 - Working on Grievance Redressal System (Webnyay), Django Python
- Worked as a Young Professional at ICAR-IGFRI, Jhansi Mar'18-May'19
- This includes the development of different software modules and applications based on organizational requirements.
- Software Developer Internship in Edureka, Bengaluru Jun'17-Feb'18
- Worked on learning management system (LMS). Most of the work is on the CakePHP MVC framework and other web technologies

REVIEWER

- Conference MIGARS, at IIST Thiruvananthapuram 2026
- Conference InGARSS, at NIT Goa ([link](#)) 2024
- Journal Advances in Space Research, Elsevier ([link](#)) 2026
- Journal Applied Soft Computing, Elsevier ([link](#)) 2026
- Journal Pattern Recognition, Elsevier ([link](#)) 2026

EXAM QUALIFIED

- GATE Qualified 2017 & 2019
- UGC-NET Qualified Jul 2026

TRAINING, CERTIFICATIONS AND WORKSHOPS

- NISAR [workshop](#) from ISRO-IIST 2025
- Multispectral Drone [pilot](#) 2024
- Legacy Responsive Web design V8 [FCC](#) 2023
- Data Analysis with Python, [FCC](#) 2023
- AICTE (ATAL FDP) training [certificate](#) on Bio-medical instrumentation 2020
- [First Position](#) at Innovation Idea Contest organized at AKTU 2020
- [Award](#) of Best Team from ICAR-IGFRI on Developing Mobile App 2018
- Certificate in [Data Analytics](#), Introductory Course in [Python](#), [OOP in C++](#) 2017
- **Research Paper presentation at International Conference in:**
 - [IGARSS](#): Brisbane Australia 2025
 - [InGARSS](#): NIT Goa 2024
 - [MIGARS](#): VCE Hyderabad 2023
 - [IWOB](#): Costa Rica 2020
 - [ICRTAC](#): VIT Chennai 2020
 - [FRSM](#): NIT Silchar 2020

PUBLICATIONS

Journal papers:

- **Manoj Kaushik**, Srinija Enimireddy, Rama Rao Nidamanuri, "*SpectraGeni: A Deep Learning Convolutional Conditional Variational Autoencoder for Synthetic Hyperspectral Data Generation from Highly Imbalanced Spectral Measurements*" <https://doi.org/10.1016/j.asoc.2026.115578>, Applied Soft Computing (2026). [Q1 Ranking Journal](#)
- **Manoj Kaushik**, Lt. Col. Jarmal Singh, Rama Rao Nidamanuri, "*Machine Learning-based Estimation and Model Transferability of Soil Organic Carbon using Ground, Airborne and Satellite Hyperspectral Remote Sensing*", <https://doi.org/10.1016/j.asr.2026.03.093>, Advances in Space Research (2026). [Q1 Ranking Journal](#)
- **Manoj Kaushik**, Rama Rao Nidamanuri, Aparna B, "*Hyperspectral Discrimination of Vegetable Crops Grown Under Organic and Conventional Cultivation Practices: A Machine Learning Approach*", <https://doi.org/10.1038/s41598-024-78714-7>, Scientific Reports 15.1 (2025): 7897. [Q1 Ranking Journal](#)
- Juwairiya Siraj Khan, **Manoj Kaushik**, Anushka Chaurasia, Malay Kishore Dutta, and Radim Burget. "*Cardi-Net: A deep neural network for classifying cardiac disease using phonocardiogram signal*." Computer Methods and Programs in Biomedicine 219 (2022): 106727, <https://doi.org/10.1016/j.cmpb.2022.106727>, [Q1 Ranking Journal](#)
- **Manoj Kaushik**, Rakesh Chandra Joshi, Atar Singh Kushwah, Maneesh Kumar Gupta Monish Banerjee Radim Burget Malay Kishore Dutta, "*Cytokine Gene Variants and Socio-Demographic Characteristics as Predictors of Cervical Cancer: A Machine Learning Approach*" Computers in Biology & Medicine, <https://doi.org/10.1016/j.compbimed.2021.104559>, 2021, [Q1 Ranking Journal](#)
- **Manoj Kaushik**, Neeraj Baghel, Radim Burget, Carlos M. Travieso, M.K.Dutta, "*SLINet: Dysphasia Detection in Children using Deep Neural Network*" Biomedical Signal Processing and Control, Volume 68, July 2021, 102798. <https://doi.org/10.1016/j.bspc.2021.102798>, [Q1 Ranking Journal](#)
- Rakesh Chandra Joshi, **Manoj Kaushik**, Malay Kishore Dutta, Ashish Srivastava & Nandlal Choudhary, "*VirLeafNet: Automatic Analysis and Viral Disease Diagnosis Using Deep-Learning in Vigna Mungo Plant*" Ecological Informatics, <https://doi.org/10.1016/j.ecoinf.2020.101197>, 2020, [Q1 Ranking Journal](#)

International Conferences:

- **Manoj Kaushik**, Jarmal Singh, Rama Rao Nidamanuri, "*Estimation of soil organic carbon content using ground, airborne, and satellite hyperspectral imagery*", presented in IEEE IGARSS 2025, Brisbane Australia
- **Manoj Kaushik**, Anagha S Sarma, Harsha Chandra, et al., "*Automated mapping of vegetable crop farms in high-resolution drone multispectral imagery using ensemble machine learning modelling*", presented in 2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2024) at NIT Goa
- Srinija, **Manoj Kaushik**, and Rama Rao Nidamanuri, "*Hybrid ensemble learning and probabilistic simulation modelling for drone-based multispectral imagery classification for crop mapping*", presented in 2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2024) at NIT Goa
- **Kaushik, Manoj**, Rama Rao Nidamanuri, B. Aparna, and A. M. Ramiya. "*Spectral discrimination of vegetable crops using in situ hyperspectral data and reference to organic vegetables*." In 2023 International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing (MIGARS), vol. 1, pp. 1-4. IEEE, 2023, <https://doi.org/10.1109/MIGARS57353.2023.10064553>
- **Manoj Kaushik**, Anagha S Sarma, Rama Rao Nidamanuri. "*CloudSegnet: A Deep Learning Based Segmentation Method for Cloud Detection in Multispectral Satellite Imagery*" in IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium, <https://doi.org/10.1109/IGARSS52108.2023.10282395>

- H Chaitra, Manohar Kumar CVSS, **Manoj Kaushik**, RG Sharathchandra, Rama Rao Nidamanuri. "**Hyperspectral Detection of Fusarium Wilt in Tomato Plants Using Machine Learning-Based Approaches**" in IGARSS 2023-2023 IEEE International Geoscience and Remote Sensing Symposium, <https://doi.org/10.1109/IGARSS52108.2023.10282890>
- **Kaushik, Manoj**, Divyanshu Singh, Malay Kishore Dutta, and Carlos Manuel Travieso González. "A deep learning approach for epilepsy seizure detection using EEG signals." Tecnología en Marcha 35, no. 4 (2022): 110-118. <https://dialnet.unirioja.es/servlet/articulo?codigo=8828180>
- **Kaushik M.**, Rani S., Yadav V. (2021) **Vocalist Identification in Audio Songs Using Convolutional Neural Network**. In: Biswas A., Wennekes E., Hong TP., Wiczorkowska A. (eds) Advances in Speech and Music Technology. Advances in Intelligent Systems and Computing, vol 1320. Springer, Singapore. https://doi.org/10.1007/978-981-33-6881-1_9
- Rani S., **Kaushik M.**, Yadav V. (2022) **Identifying Mood in Music Using Deep Learning**. In: Raje R.R., Hussain F., Kannan R.J. (eds) Artificial Intelligence and Technologies. Lecture Notes in Electrical Engineering, vol 806. Springer, Singapore. https://doi.org/10.1007/978-981-16-6448-9_55

FUNDING and GRANTS

- ANRF(SERB) ITS grant for IGARSS, Brisbane Australia, 2025
- CSIR foreign travel grant for IGARSS, Brisbane Australia, 2025
- IIST international travel grant, 2025
- IEEE IGARSS Travel Grant, Brisbane Australia, 2024
- IEEE InGARSS Travel Grant, NIT Goa, 2024
- IEEE IGARSS Travel Grant, California, USA, 2024

OTHER PROJECTS

[Webnyay: an online dispute resolution system](#)

- Client project of Thotnr Pvt. Ltd., Webnyay is an online grievance redressal system built on top of Django, ReactJS, PostgreSQL, and integrated with advanced AI documentation

[LMS \(Learning Management System\) modules for Edureka pvt. Ltd.](#)

- PHP, Angularjs based LMS system.

[Iron corrosion image Segmentation using Deep Learning](#)

- Segment out various colored corrosion parts in Iron using UNET Deep Learning Architecture

[Mailing Management System: A PHP Web App](#)

- Application to send email notifications to employees to avoid the penalty for late settlements of their office Advances

[Chara-App: Hybrid Mobile Application](#)

- Mobile application provides all information and technologies regarding better Fodder production for farmers

[Drishticone: College Newsletter Website](#)

- Web portal for college students in which college-related news and placement guidance-related articles can be found easily in one place

[Analysis of Parallel Algorithms](#)

- Analyzed the time and space complexity of parallel algorithms over sequential algorithms. Designed a Parallel hybrid sort algorithm to run on multiple cores using MATLAB (PCT tool)

RESPONSIBILITY AND EXTRACURRICULAR

- Young Professional Activity Chairs in Conference MIGARS 2026 [link](#)
- GEHU Bhimtal MOU with ARIES Nainital on research projects
- Volunteered in International Yoga Mohotsav in IIST Thiruvananthapuram.
- Managed, taught, and volunteered in DST-funded three-week winter school training in IIST
- Managed, taught, and volunteered in IEEE GRSS's one-day hands-on workshop in IIST
- Managed and volunteered Geo Innovation challenge in April'2022 organized by DST, Govt. of India
- Managed and volunteered in IC3A2020 Organized by AKTU and CAS in February'2020

TECHNICAL SKILL

Programming Languages and Technologies: Python, TensorFlow, OpenCV, Keras, C, C++, SQL

Area of Interest: Hyperspectral Remote Sensing, Artificial Intelligence, Data Science, Data Analytics and Development

Platforms: Linux, Windows, Google Colab

Tools: PyCharm, Anaconda, Nvidia CUDA, LaTeX, MS Office, Dev C++, and Adobe Photoshop